

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Technology (CSE)

QUIZ #2 (SET-A)

DURATION: 30 MINUTES

SUMMER SEMESTER, 2023-2024

FULL MARKS: 15

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

1. You are a newly appointed zookeeper, and your first task is to find rooms for all the animals: Iguana (I), Jaguar (J), and Koala (K). Each animal can be assigned into one of the four rooms: North room (N), East room (E), South room (S), and West room (W), while following the given constraints:

- The Jaguar can not share a room with any other animals.
- The Iguana and Koala must be in different rooms.
- The Koala can only be in the East room or the South room.

- a) Formulate the problem as a CSP by identifying the variables, their domains, and the constraints.

3
(CO1)
(PO1)

Solution:

Variables: $\{I, J, K\}$, **Domains:** $I \in \{N, S, E, W\}$, $J \in \{N, S, E, W\}$, $K \in \{N, S, E, W\}$,
Constraints:

- $J \neq I, J \neq K$
- $K \neq I$
- $K \neq \{W, N\}$

Rubric:

- 1 point for correctly stating the variables.
- 1 point for correctly stating the domains.
- 1 point for correctly stating the constraints.

- b) Suppose you have done some filtering and the following values remain in each variable's domain: $I = \{E, W\}$, $J = \{S\}$, $K = \{E\}$. List all of the arcs and show if they are consistent or not?

6
(CO1)
(PO1)

Solution:

Only the $I \rightarrow K$ arc is not consistent, all others are consistent.

Rubric:

- 1 point for correctly identifying the consistency of each arc. There are 6 arcs in total.

2. Consider a CSP with $L + R$ variables. There are d possible assignments to each variable. The CSP only has binary constraints. If we remove one of the binary constraints, then the CSP can be decomposed into two independent subproblems: A *left* CSP (V_L) with L variables and a *right* CSP (V_R) with R variables. Let, $X \in V_L$ and $Y \in V_R$ be the variables that are connected by the only constraint that joins V_R and V_L . 6
(CO1)
(PO1)

Without removing any constraints, we assign a value to one of the variables in V_L and run forward checking. What is the **maximum** number of variables whose domains could be changed as a result of this?

Solution:

The domains will only change if the variable is connected to the variable that is being changed. A variable on the left side could be connected to at most $L - 1$ variables belonging to V_L in addition to the 1 variable belonging to V_R . So the most number of variables that could be affected is L .

Rubric:

- 3 points for each correct answer.